

Franklin Wohnoutka

Minneapolis, MN | Franklinw1023@gmail.com | 651-399-6868 | [Portfolio](#)

EDUCATION

Drake University, Des Moines, IA

GPA: 3.82 | Dean's List | Expected Graduation: May 2027

Bachelor of Science in Data Analytics, Mathematics, Computer Science

Minor in Physics

TECHNICAL SKILLS

Languages: Python, Java, SQL, R, C, JavaScript, NetLogo

Frameworks: PyTorch, TensorFlow (AlexNet, ResNet, VGGNet), React, React Native

Tools & Software: Git, Microsoft Office, MySQL, AWS (EC2, S3, RDS, Flask, Kinesis), Azure, Supabase, Expo, EAS

Concepts: Object-Oriented Programming, Machine Learning, Database Design, Data Visualization, Multivariate Calculus

RELEVANT EXPERIENCE

Tutor

Drake University Mathematics & Computer Science Tutoring Lab, Des Moines, IA

November/2025 – Present

- Provide lab style tutoring to students in a range of Mathematics and Computer Science courses
- Support student understanding through problem solving strategies and examples rather than giving direct answers
- Update professors on student progress and identify areas where additional support may be needed

Research Assistant

M.I.S.S.F.I.T (Magno-Ionized Space Shield for Interplanetary Travel), Des Moines, IA

May/2025 - Present

- Code and run simulations for student-based NASA funded research project run through Drake University
- Utilize Git for version control and collaborative development within a team environment
- Analyze and graph simulation results to present progress at weekly meetings
- Collect and process data on different heavy nuclei from CRIS and ACE satellite data repositories

PROJECTS

Posted: Fits & Hauls – Fashion Social Media App (iOS) | App Store: [Posted-Link](#)

June/2026- Present

- Independently designed and shipped a full-stack fashion sharing iOS app to the App Store, building a front end with React Native/Expo, a backend with Supabase/PostgreSQL and managing native development with EAS
- Developed a PostgreSQL backend with row-level security policies, real-time subscriptions, and Postgres RPC functions for features such as friend discovery, follow requests, and streak resets
- Built core features including a social feed, wardrobe calendar, clothes tagging system, and theme customization

Random Search for Treasure on Connected Graphs – Mathematics Capstone

March/2026-May/2026

- Developed a graph searching method, inspired by “cops and robbers”, on a connected graph of n vertices
- Built a recursive Python simulation to compute capture time and proved that $\Pr(\text{capt}(H)=k)=1/n$ for a hunter H on any connected graph with n vertices

Immigration Enforcement Simulation using NetLogo

January/2026

- Created a sophisticated simulation in net logo modeling fear and crime rates in communities
- Executed extensive data analysis on model and presented findings

Machine Learning Image Categorization

November/2024-December/2024

- Trained AlexNet, ResNet, and VGGNet models on a dataset of 1352 images of birds (25 different species)
- Used data normalization and fine-tuning with a confusion matrix achieving 98.4% accuracy with AlexNet

EXTRACURRICULAR ACTIVITIES

Secretary, Society of Physics Students, Des Moines, IA

February/2025- Present

Mentee, Vermeer Digital Bulldogs, Des Moines, IA

January/2025 – Present

Member, Wallin Scholars, Minneapolis MN

August/2023-Present